EZHUMIN TECHNOLOGIES

MINOVA
Product Portfolio

SMART HYBRID ENERGY SYSTEMS

OFFICIAL TECHNICAL PORTFOLIO • **EZHUMIN TECHNOLOGIES** • GLOBAL EDITION

About Company

EZHUMIN TECHNOLOGIES is an advanced clean-technology pioneer specializing in next-generation hybrid renewable energy infrastructure. Engineered to address the compounding inefficiencies of standalone renewable sources, our core engineering ethos converges solar PV, micro-wind kinetics, utility-grade energy storage, and proprietary AI-driven optimization matrices.

VISION

To architect the foundational framework for decentralized global energy self-sufficiency, displacing conventional baseload assets with smart, non-intermittent hybrid technologies.

MISSION

To engineer, manufacture, and scale high-performance multi-source renewable microgrids that deliver resilient, continuous, and economically optimized power to all sectors.

CORE VALUES

Uncompromising engineering integrity, structural longevity, grid-interactive resilience, and a steadfast dedication to absolute technological sustainability.

MADE IN INDIA, FOR THE WORLD

Proudly researched, developed, and precision-manufactured within India. Our architecture conforms strictly to international IEC and IEEE compliance baselines, serving as a future-proof global brand.

CORE TECHNOLOGY VECTORS

AI-Powered Energy Systems: Standard setups suffer from intermittent production. MINOVA uses dynamic logic arrays to analyze weather, cycle health, and load draw, predicting and routing currents automatically to achieve zero-drop operations.

Sustainable Lifecycles: From non-toxic LFP battery configurations to high-recyclability turbine composites, every structural layer is chosen to maximize operating life and minimize net carbon debt.

Why MINOVA

Traditional clean energy architectures isolate solar and wind modalities, leading to significant grid vulnerability and multi-hour generation shortfalls. MINOVA redefines infrastructure resilience by consolidating co-located kinetic and photovoltaic generation channels into a unified physical asset.

Solar + Wind Co-Generation

Captures complementary diurnal cycles. Photovoltaic yields peak mid-day, while aerodynamic kinetic systems leverage high thermal nocturnal winds, guaranteeing a continuous multi-phase generation profile.

Integrated Battery Storage (BESS)

Utilizes high-density Lithium Iron Phosphate (LiFePO4) chemistries managed by multi-tier active balancers to buffer structural transient spikes and extend runtime safety thresholds.

AI Energy Management Engine

Executes complex predictive algorithmic dispatching, load shaping, and power-factor leveling to preserve local thermal and physical component lifecycles.

IoT Monitoring & Diagnostics

Continuous telemetry reporting utilizing dual-band WiFi and cellular fallback channels, broadcasting detailed sensor array telemetry to our global operation cloud center.

Remote Diagnostics & Control

Enables OTA code patching, isolated line disconnects, thermal mitigation switches, and programmatic safety override execution without physical service interventions.

Modular & Scalable Architecture

Hot-swappable structural racking, parallel power control topologies, and standardized mechanical interfaces allow seamless field expansion up to multi-megawatt capacities.

Product Categories

The MINOVA ecosystem encompasses a comprehensive portfolio designed across multiple power tiers to satisfy specific spatial, operational, and financial parameters.

MINOVA L Series

STREET LIGHTING SYSTEMS

Autonomous hybrid municipal and industrial highway illumination assets featuring integrated generation arrays.

MINOVA H Series

RESIDENTIAL SYSTEMS

Premium micro-hybrid configurations optimized for high power quality and residential grid independence.

MINOVA A Series

AGRICULTURE SYSTEMS

High-torque off-grid agricultural micro-grids for heavy-duty pumping and continuous field machinery loads.

MINOVA C Series

COMMERCIAL SYSTEMS

Mid-tier commercial power architectures engineered for peak-shaving and facility carbon minimization.

MINOVA I Series

INDUSTRIAL SYSTEMS

Heavy multi-megawatt industrial micro-grids featuring rigorous sub-millisecond switching dynamics.

MINOVA B Series

BATTERY STORAGE

Scalable LiFePO4 rack enclosures with advanced liquid cooling and multi-point battery protection circuits.

MINOVA Core

ENERGY MANAGEMENT CONTROLLER

The central computational hardware executing multi-source MPPT algorithms and local automation rules.

MINOVA Connect

MOBILE APPLICATION & CLOUD

Real-time interaction matrix providing enterprise asset tracking, diagnostic alerts, and performance metrics.

MINOVA L Series | Street Lighting

Completely autonomous, rugged municipal infrastructure illumination platforms. Combining localized vertical aerodynamic micro-turbines and high-efficiency bifacial monocrystalline solar cells to guarantee operational performance across adverse climatic conditions.

MODEL	LUMENS / POWER	BATTERY CAPACITY	APPLICATION VECTOR	DISTINCTIVE ENGINEERING FEATURES
MINOVA L50	6,500 lm / 50W	400 Wh LiFePO4	Urban Footpaths / Parks	Integrated passive PIR motion profiling, ultra-compact profile
MINOVA L100	13,000 lm / 100W	800 Wh LiFePO4	Collector Roads / Residential	Dual-axis structural mounting options, low wind cut-in
MINOVA L150	19,500 lm / 150W	1,200 Wh LiFePO4	Arterial Roadways	Bifacial solar surface tracking, reinforced marine-grade alloy
MINOVA L200	26,000 lm / 200W	1,600 Wh LiFePO4	Highways / Expressways	External wind aerodynamic fairings, real-time node telemetry
MINOVA L300	39,000 lm / 300W	2,400 Wh LiFePO4	Industrial Logistics Yards	Dual-luminaire modules, high-altitude structural rating

MINOVA H Series | Residential

Premium grid-interactive and off-grid micro-hybrid generation centers. Engineered to isolate residential consumers from frequent brownouts, high tariff escalations, and power quality anomalies through sub-millisecond line switching capabilities.

MODEL	SOLAR ARRAY CAP	WIND TURBINE ASSET	DAILY AVG YIELD	BESS STORAGE OPTION
MINOVA H2	2.0 kWp	0.5 kW Vertical Axis	10 - 12 kWh	5.0 kWh Modular Rack
MINOVA H3	3.0 kWp	1.0 kW Vertical Axis	16 - 20 kWh	5.0 / 10.0 kWh Expandable
MINOVA H5	5.0 kWp	1.5 kW Horizontal Axis	28 - 34 kWh	10.0 kWh High Output
MINOVA H7	7.5 kWp	2.0 kW Horizontal Axis	42 - 50 kWh	15.0 kWh High Output
MINOVA H10	10.0 kWp	3.0 kW Dual Turbines	58 - 70 kWh	20.0 kWh Enterprise Base

Commercial & Agriculture Systems

MINOVA A SERIES | AGRICULTURE SYSTEMS

Built for harsh conditions and high transient starting currents. Optimized for multi-phase agricultural pumps, automated micro-irrigation processing, and rural off-grid operations.

MODEL	PUMP POWER RATING	HYBRID GEN CAPACITY	DAILY WATER DELIVERY OUTPUT (100M HEAD)
MINOVA A3	3.0 HP Pumps	4.5 kW Combined	Up to 45,000 Liters / Day
MINOVA A5	5.0 HP Pumps	7.5 kW Combined	Up to 75,000 Liters / Day
MINOVA A10	10.0 HP Pumps	15.0 kW Combined	Up to 160,000 Liters / Day
MINOVA A20	20.0 HP Pumps	30.0 kW Combined	Up to 330,000 Liters / Day

MINOVA C SERIES | COMMERCIAL SYSTEMS

Designed for corporate campuses, office parks, and commercial operations. Engineered for seamless peak-tariff mitigation, demand-charge reduction, and automated power factor correction.

MODEL	INVERTER RATING	SOLAR CONFIGURATION	WIND KINETIC MATRIX
MINOVA C10	10 kW Multi-Phase	12 kWp Monocrystalline	3 kW Aerodynamic Wind Array
MINOVA C20	20 kW Multi-Phase	25 kWp Monocrystalline	5 kW Aerodynamic Wind Array
MINOVA C30	30 kW Multi-Phase	35 kWp Monocrystalline	10 kW Aerodynamic Wind Array

MODEL	INVERTER RATING	SOLAR CONFIGURATION	WIND KINETIC MATRIX
MINOVA C50	50 kW Multi-Phase	60 kWp High Efficiency	15 kW Aerodynamic Wind Array

MINOVA I Series | Industrial Systems

High-capacity, heavy-duty microgrid installations engineered specifically for automated production environments, heavy manufacturing facilities, and localized grid reinforcement configurations.

MODEL	CONTINUOUS AC OUTPUT	SOLAR PV MATRIX MAX	WIND TURBINE ASSEMBLY	ENCLOSURE ARCHITECTURE
MINOVA I25	25 kW	30 kWp	10 kW Turbine	NEMA 4X Multi-Cabinet
MINOVA I50	50 kW	60 kWp	20 kW Turbine	NEMA 4X Multi-Cabinet
MINOVA I100	100 kW	120 kWp	40 kW Multi-Turbine	IP65 Thermal Enclosure
MINOVA I250	250 kW	300 kWp	100 kW Multi-Turbine	10ft ISO Climate Container
MINOVA I500	500 kW	600 kWp	200 kW Multi-Turbine	20ft ISO Climate Container
MINOVA I1000	1000 kW (1 MW)	1200 kWp	400 kW Multi-Turbine	40ft ISO Climate Container

MINOVA B Series | Battery Storage

Utility-grade Lithium Iron Phosphate (LiFePO4) Battery Energy Storage Systems (BESS). Equipped with sub-millisecond multi-point voltage telemetry, integrated active balancing circuitry, and multi-stage fire mitigation technologies.

MODEL	NOMINAL CAPACITY	CELL CHEMISTRY	VOLTAGE PROFILE	THERMAL CONTROL MATRIX
MINOVA B5	5.12 kWh	LiFePO4 (LFP)	51.2 VDC	Passive Extruded Aluminum
MINOVA B10	10.24 kWh	LiFePO4 (LFP)	51.2 VDC	Forced Convection Fan Array
MINOVA B15	15.36 kWh	LiFePO4 (LFP)	102.4 VDC	Forced Convection Fan Array
MINOVA B20	20.48 kWh	LiFePO4 (LFP)	204.8 VDC	Active Solid-State Refrigeration
MINOVA B50	51.20 kWh	LiFePO4 (LFP)	409.6 VDC	Liquid-Glycol Circulation Loops

Smart Platform Architecture

The MINOVA ecosystem is coordinated by a powerful multi-tier automation layer that balances power conversion efficiency with cloud diagnostics.

MINOVA CORE / CORE+

Our edge hardware controller featuring dual-core processors to optimize real-time multi-source MPPT algorithms at a 20kHz sampling frequency.

MINOVA EDGE

Locally deployed software stack that executes high-speed islanding protection and handles transient load surges without requiring cloud connectivity.

MINOVA SYNC

A hardware synchronization matrix that dynamically aligns inverter output phases with external grid waveforms to ensure clean transitions.

MINOVA CONNECT APP

User application delivering real-time telemetry updates, detailed battery diagnostic breakdowns, and customizable charge scheduling options.

MINOVA CLOUD & AI

An enterprise-grade IoT platform that leverages machine learning to forecast localized generation capacity and optimize system lifecycle efficiency.

Technology Architecture

The systematic structural flow within a MINOVA smart hybrid installation routes harvested power seamlessly through high-efficiency conversion stages to ensure consistent output quality.

Solar Panels (Photovoltaic Conversion Array)

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Wind Turbine (Kinetic Aerodynamic Array)

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Hybrid MPPT (Simultaneous Dual-Source Max Power Tracking)

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MINOVA Core AI Controller (Real-Time Local Routing Decisions)

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Battery (High-Density LiFePO4 Energy Storage System)

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Hybrid Inverter (Pure Sine Wave Multi-Phase DC-AC Inversion)

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Grid / Consumer Load (Interactive Local & Grid Destination Distribution)

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MINOVA Cloud & Connect Mobile App (Distributed IoT Management Platform)

Strategic Future Roadmap

EZHUMIN TECHNOLOGIES is committed to systematic R&D progression, scaling product execution cleanly across municipal, residential, commercial, and utility sectors globally.

2026	**Municipal Infrastructure Dominance** Rollout of the MINOVA L Series across Indian highways, urban development zones, and smart city deployments.
2027	**Residential Market Penetration** Launch of the H Series consumer microgrids alongside scalable B5/B10 residential battery packs.
2028	**Commercial Sector Expansion** Deployment of mid-tier C Series energy setups tailored for corporate complexes and agricultural cooperatives.
2029	**Heavy Industrial Deployment** Commissioning of high-capacity containerized I Series microgrids to support automated manufacturing plants.
2030	**Next-Gen AI Platform Launch** Introduction of autonomous peer-to-peer micro-grid trading networks managed entirely via MINOVA AI.
2031	**Global Market Scaling** Expanding manufacturing footprint and establishing international distribution channels across Europe, APAC, and North America.

Global Contact & Support

Partner with EZHUMIN TECHNOLOGIES to transform your energy profile. Our engineering teams provide comprehensive site analysis, hybrid architectural optimization, and turn-key deployment services worldwide.

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Disclaimer: The technical specifications, design vectors, and model configurations contained within this portfolio reflect structural baselines current at the time of publication. EZHUMIN TECHNOLOGIES reserves the right to modify mechanical layout, structural material selection, or software codebase rules in line with continuous product refinement protocols.